

Requirements Table

Name: MANASA K SRN: PES1UG25CS824

Functional Requirements (FR-001 – FR-005)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a Donor Partner to post a surplus food batch, specifying quantity, a perishable expiry countdown timer, and one or more dietary tags (e.g., vegetarian, nut-free, dairy-free).	High	Pass: A newly posted batch appears on the claim board immediately with a visible countdown timer and correct dietary tags. Fail: Batch is saved without an expiry timer or is not visible to shelters.	Core donation-intake functionality; accurate expiry and dietary data are essential for food safety and shelter decision-making.
FR-002	Functional	The system shall allow a Shelter Coordinator to search and filter available food batches by dietary tag, distance from the shelter, and remaining time before expiry.	High	Pass: Applying a filter (e.g., 'nut-free' within 5 km) returns only batches matching all selected criteria within 2 seconds. Fail: Filtered results include non-matching batches or omit matching ones.	Shelters must quickly locate suitable, nearby, still-valid food to make pickup decisions before expiry.
FR-003	Functional	The system shall allow a Shelter Coordinator to reserve an available food batch and receive a unique pickup verification code and a designated pickup window.	High	Pass: Upon reservation, the batch is removed from the claim board for other shelters and a unique verification code is generated and displayed/emailed to the coordinator. Fail: Batch remains reservable by multiple shelters simultaneously (double-booking).	Prevents double allocation of the same surplus batch and provides a verifiable handoff mechanism at pickup.
FR-004	Functional	The system shall automatically remove a food batch from the claim board once its expiry countdown timer reaches zero, regardless of reservation status.	High	Pass: A batch whose timer reaches 00:00 is marked 'Expired' and no longer appears in browse/search results within 60 seconds. Fail: An expired batch remains active and reservable on the claim board.	Protects shelters and recipients from food-safety risk by ensuring only valid, edible surplus is offered.
FR-005	Functional	The system shall allow a Donor Partner to update the quantity/expiry of, or cancel, a previously posted food batch, provided it has not yet been reserved by a shelter.	Medium	Pass: Donor edits/cancels an unreserved batch and the change is reflected on the claim board instantly. Fail: Donor is able to cancel or edit a batch that has already been reserved without notifying the reserving shelter.	Allows donors to correct posting errors or reflect real-world changes in surplus availability.

Non-Functional Requirements (NFR-001 – NFR-002)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The platform shall dispatch instant push notifications to all verified shelters within a 5 km radius when a high-quantity perishable food batch is listed.	High	Pass: Benchmarking tests confirm notification latency under 10 seconds and delivery over an authenticated, encrypted channel under simulated peak load (500 concurrent postings). Fail: Notification latency exceeds target or is sent over an unencrypted channel.	Timely, secure alerts maximize the chance that perishable surplus is claimed before it expires.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-002	Usability & Reliability	The platform shall remain usable on low-bandwidth mobile connections, with core pages (post batch, browse, reserve) loading in under 3 seconds on a simulated 3G network, and shall maintain 99% monthly uptime.	Medium	Pass: Load-testing on a throttled 3G profile shows page load < 3s for core flows, and uptime monitoring logs show $\geq 99\%$ availability over a 30-day period. Fail: Any core page exceeds 3s load time on 3G or monthly uptime falls below 99%.	Donor partners and shelter coordinators (often small NGOs) may operate in areas with limited connectivity; the platform must remain accessible.